

The Process of Strategic, Agile, Innovation Development: A Healthcare Systems Implementation Case Study

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ABSTRACT

Despite many attempts to introduce computerization in the healthcare industry, the majority of the current healthcare information systems still fail to meet the rising expectations of patients for service. This study aims to understand how agility and innovation capabilities can be strategically nurtured, developed, and managed to upgrade the quality of healthcare services. Based on a case study, a process model is developed to explain that an agile innovation strategy is a complex helix process involving a firm's sensitivity and responsiveness to integrating and reconfiguring its resources to cope with innovative change. Three key managerial contributions for IT and medical practitioners are presented.

Keywords: *Agility, Case Study, Information Technology in Healthcare, Innovative Capabilities, Innovation in Services*

1. INTRODUCTION

In today's competitive and demanding economic environment, businesses in all industries, including healthcare, are undergoing profound changes. With global expenditure of more than US\$6.5 trillion in 2012 (World Health Organization 2012), the healthcare industry is known to be the largest industry worldwide to be undergoing a process of restructuring and reengineering. It is intended to provide better patient services and care through the use of IT. According to the recent Ernst & Young

health care industry report (2014), Healthcare Information Technology (HIT), and the data it yields, serves as ballast, holding the ship steady in its journey forward. It further suggested that health care organizations must continue to invest significant capital — human resources, time and money — in information systems design, implementation and training (Ernst & Young 2014). Recognizing the scale and continued growth of this industry, along with its potential to transform societal health and wellness levels, Singapore's Ministry of Health has a mission to develop the world's most cost-effective health-

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care system to ensure that Singaporeans—the country's key resources—have an ideal state of health (World Health Organization 2007).

Singaporean hospital-industry leaders are seeking ways to explore the opportunities and challenges faced in the development of medical informatics, with minimal guidance or policy directives from the government. Healthcare experts, policymakers, and consumers find that computerization and the automation of health records and health-related processes can critically transform the nature of their industry (Dwivedi et al. 2007; Heathfield et al. 1999). To shape its future, not only must an organization be increasingly agile to detect market changes and opportunities (Sambamurthy et al. 2003) for significantly greater flexibility and responsiveness (Gould 1997), they must also be able to innovate (Lawson et al. 2001). Hence, agility is critical for organizations to sense and respond readily and effectively to future needs (Overby et al. 2006).

The potential for information technology (IT) to enable enterprise agility has received increasing attention from practitioners and academics in recent years (Mathiassen et al. 2006; Overby et al. 2006; Sambamurthy et al. 2003; van Oosterhout et al. 2006). However, despite advances in knowledge about the relationship between IT and organizational efforts to achieve agility, enterprise agility is often treated as a “black box”; empirical validation of this relationship remains rare (e.g. Overby et al. 2006; Seo et al. 2008). The importance of agility in a number of sectors, including healthcare, has been largely ignored. In view of this, agility theory was employed as the theoretical lens to structure this case study of an agile and highly innovative Singaporean hospital.

The case study was conducted with a private hospital that successfully designed and injected agile and innovative philosophies into its healthcare system. This hospital had received a record high level of patient complaints and did not have a fine IT system before its changes to transform itself for greater agility and innovation. It developed various agile and innovative capabilities, while at the same time creating a

health information system (HIS) which suited its strategic needs. In each phase, the agile and innovative capabilities directly contributed to the achievement of its strategic goals in upgrading its healthcare services, and it also facilitated the HIS implementation. Through this parallel process, the development of agile and innovative capabilities and the new HIS jointly contributed to the hospital's success in transforming a record high level of patient complaints to becoming the best hospital in the Singaporean Ministry of Health Patient Satisfaction Surveys from 2004 to 2008.

Agile and innovative transformation motivated this study. This article addresses a research gap by conceptualizing the agility-innovation capability-development process in delivering a strategic information systems implementation in a hospital setting.

2. THEORETICAL BACKGROUND

Today, the demanding nature of patients requires the transformation of the healthcare industry in order to fulfill increasing expectations. One important way to improve healthcare services is to upgrade the service system through the use of advanced information technology (IT) or HIS (Abraham et al. 2011; Heeks 2006; Holden et al. 2010; Lee et al. 2011; Lluch 2011). Unfortunately, the healthcare community is making slow progress in meeting these demands (Lee et al. 2011). Compared to other healthcare technologies, HIS has been slow in its adoption (Konstantinidis et al. 2010).

A HIS is introduced because of a hospital's needs to manage business-related data through a centralized system (Nemeth et al. 2005). A HIS consists of various IT applications which may include different modules such as registration, and electronic medical records and laboratory systems (Mantzana et al. 2007). HIS-related research often refers to a multidisciplinary body of knowledge linked to the design, development, implementation,

and use of information-intensive technologies in healthcare settings (Chiasson et al. 2004).

We begin with IT in the healthcare context, noting the differences between IT in business organizations and IT in healthcare. We then introduce the concept of enterprise agility. Finally, we explore theories on innovative capabilities.

2.1. IT in the Healthcare Context

Over the years, a body of HIS research developed based on IT adoption and (or) diffusion theories (e.g. Moore et al. 1991; Rogers 1995) to focus on the characteristics of technology and on users' perceptions of it. Building on the theories of innovation diffusion, the literature is mainly focused on the adoption of healthcare systems based on the Technology Acceptance Model (TAM) (Holden et al. 2010; Melas et al. 2011), and the influence of the fit between technology and the task of its adoption (Goodhue et al. 1995; Zigurs et al. 1998). For example Kijisanayotin et al. (2009) applied the Unified Theory of Acceptance and Use of Technology (UTAUT) structural model to investigate factors that influenced IT adoption in community health centers in Thailand. They validated the UTAUT model in the field in a context of a developing country's healthcare system. Although many insightful research findings were generated from such research, there are two noteworthy limitations. First, a cross-section analysis of how technology characteristics and users' perceptions of such characteristics affect IT innovation adoption provides many insightful findings. However, IT innovation has also been recognized as a process of movement, that is, of transforming inventions into usable forms for practical use (King et al. 1994); this should not be ignored. Recognizing and investigating the evolutionary nature of the introduction of IT-based innovation would yield more detailed and practical prescriptions for other healthcare organizations to follow. Second, HIS adoption focuses on delivering services to patients according to one of the seven principles of Klein and Myers (1999)—namely, the principle of contextualization. This suggests that a key

research task is to seek meaning in context; the adoption practices could be quite different from other industries' system-implementation strategies (Chiasson et al. 2004). For example drawing on field data collected in the Indian public healthcare sector, Noir and Walsham (2007) challenged the simplistic view that implementing HIS will translate directly into the intended efficiency gains. Instead, they advocated a theoretical position that takes seriously the interaction between the environment and the organizational field.

In a recent review of HIS development, it appears that innovation issues can be solved with continual improvement through the incorporation of the "design-reality gap" concept (Heeks 2006) for new products and services. Extending this concept here, we explore how a Singaporean hospital designed its IT systems based on the "kaizen" model of continual improvement. The aim is to address the design-reality gap. In particular, the goal is to understand the key agile and innovative capabilities that this hospital strategically developed to achieve its agile HIS.

2.2. Enterprise Agility

The "agility" of an enterprise refers to a capacity within an organization to react to changes in its environment, act on them, and use them to its profit (Ganguly et al. 2009; Huuomonen 2011). This critically depends on its resources and capabilities (Nambisan 2002), including innovative capabilities (Malone et al. 1999). Agility is vital to the innovation and competitive performance of firms in contemporary business environments (Sambamurthy et al. 2003). The term "enterprise agility" is based on the posited historical progression from mass manufacturing to agile manufacturing (Bottani 2010; Dove 2001; Ganguly et al. 2009; Sharp et al. 1999). It was then adopted by the software-development industry (Cockburn 2001). The emergence of the concept is rooted in the new business era in which "change" is a major characteristic (Sharifi et al. 1999). Thus, an agile enterprise has the capability to survive and prosper in a competitive environment of continuous and unpredict-

able change, by spontaneously reacting to the market (Gunasekaran 1999). This definition is further refined by van Overby et al. (2006) as the ability to sense environmental change and to respond readily to that changing environment with improved productivity (Mathiassen et al. 2006), through the innovative use of a firm's resources and capabilities.

An agile enterprise is not only capable of successfully implementing change, it is nimble and can respond quickly and efficiently to both expected and unexpected events in its environment (Mathiassen et al. 2006), making use of its capabilities. Such capabilities relate to customers (Nambisan 2002), internal operations for innovation and competitive action (Malone et al. 1999), and the utilization of networks to explore opportunities for innovation and competitive action (Choudhury et al. 1999). To compete in a context of high standards of performance, firms must be able to develop more complex bases of resources and capabilities innovatively (Ferrier et al. 1999; Sambamurthy et al. 2003)—an assertion which has continued to receive attention for several decades within industry and academia (Lyytinen et al. 2006). In addition, advances in information technology (IT) have allowed businesses to automate the design, sales, and support of their products and services and have given enterprises the ability to sense and respond to changes in a competitive environment (Gallagher et al. 2008; Lior 2011; Tallon et al. 2011). As for information-system development to enable agility, an organization needs to sense and respond to environmental change. As noted by Lyytinen and Rose (2006), the process of sensing and responding to the components of agility is associated with two learning abilities: exploration and exploitation. The former is related to how an organization acquires the capabilities to sense IT innovation; the latter is about how they systematically and continually improve their response capability.

However, despite a growing body of research in this area, a number of gaps have been identified in the literature. First, the development of agility is not a straightforward process. In fact, it often involves an evolutionary

process comprising distinct phases with different focuses (Tan et al. 2009). Thus, it would be valuable from a process viewpoint to identify prescriptions for attaining enterprise agility through different phases (Chen et al. 2010) in order to understand the different innovation activities and focuses from the IS-development perspective. Second, there are two main streams within agility studies—one focuses on IT diffusion and agile adoption of practices as a means to achieve enterprise agility (e.g. Hovorka et al. 2006); the other offers prescriptions for attaining enterprise agility through IT (e.g. Overby et al. 2006; van Oosterhout et al. 2006). These streams have barely considered how innovative capabilities influence agile IT development. Here, we investigate these identified gaps.

2.3. Innovative Capabilities

“Innovative capabilities” refer to a firm's ability to develop new products, services, and (or) markets by aligning strategic innovative behaviors and processes to achieve their usual solutions and also novel solutions (Wang et al. 2007). In general, there are two types of innovative capabilities: incremental and radical (Dewar et al. 1986). Incremental innovative capability is a capability to generate innovation in a progressive manner, which refines and reinforces existing products and services. Radical innovative capability is a capability to generate and transform existing products and services significantly (Subramaniam et al. 2005). Fundamentally, the distinction between incremental and radical innovative capabilities lies in the ability to draw on organizational knowledge. No doubt such identified capabilities are particularly associated with enhancing a firm's performance (D'Este 2002; Deeds et al. 1999; Pan et al. 2007), evolution, and survival in the context of external competition and change (Deeds et al. 1999; Lazonick et al. 2005). The challenge remains that an acceptable, comprehensive, and systematic understanding of innovative capabilities has yet to be established (Lawson et al. 2001). More importantly, the scarcity of research on innovative change

in the healthcare industry (Amble et al. 1998; Aranda et al. 2002; Martin et al. 1993) underpins the value of our research.

To innovate, an organization must develop new products, services, and (or) markets with continual and incremental learning and experimenting, both within and outside the organization (Aranda et al. 2002). Here, we refer to learning as the ability to acquire, assimilate, transform, and exploit existing knowledge, and to integrate and generate new knowledge (Aranda et al. 2002; Teece et al. 1997). We recognize that knowledge and the transfer of learning are of the utmost importance to the implementation of system innovation (Metcalfe 1995).

Integration and coordination have been viewed as important elements that facilitate the deployment of resource reconfiguration (Teece et al. 1997). These two elements can determine how different resources are integrated to create new ways of performing new sets of activities—for example by transforming existing resources into new resources that could potentially be used to respond to volatile environments (Teece et al. 1997). A successful innovation requires recognition of the new and unfamiliar capabilities of IT in rethinking business processes, instead of relying on existing capabilities (Whitman 1996). Unfortunately, most organizations do not always provide sufficient supportive resources for their employees who are likely to engage in IT innovation (Wu 2003). In view of the identified research gaps, our study explores the development of an acceptable, comprehensive, and systematic framework for successful innovation (Lawson et al. 2001).

3. METHODOLOGY

We chose a case-study approach to investigate the introduction of agility and innovation in the Singaporean health sector through IT adoption. This approach provides us with a means to explore contemporary events in the case company (Winter 2003) through an empirical research focus on “how” and “why”

(Lazonick et al. 2005). The main objective is to discover how a service firm—in this case, the Alexandra Hospital (AH)—innovated in a volatile environment using IT to establish a new industry standard for Singaporean healthcare. In addition, the scarcity of research on this phenomenon (Amble et al. 1998; Aranda et al. 2002; Martin et al. 1993) supports our choice of this methodology. In particular, an interpretive case study is seen as an appropriate method of empirical inquiry (Lazonick et al. 2005) when the studied phenomena are complex and not easily separated from their original context (Yin 1994). Thus, from an interpretive perspective, we use a model of capability-development in achieving agile innovation strategies. We draw on the rich insights available—in this case by examining the major events of the entire innovative response process in the context of the volatile Singaporean hospital environment.

The case study of AH is particularly appropriate for this research. It has been cited by the Singapore Health Minister as a key prototyping center, which is leading and setting future standards for the Singapore healthcare industry¹. In particular, we appraise its creativity in exploring available IT, its agility in responding to new ideas, and its innovation in designing a HIS to formulate a new service model for the Singapore healthcare industry.

3.1. Data Collection

The case study was conducted over a period of six months and involved 24 face-to-face interviews with project group members recommended by the AH Director of Projects and Operations. Most interviews were tape recorded and transcribed, with photographs and additional notes taken to collect the best possible data on the features of the case. Each interview lasted between one and a half and three hours, with informants including nurses, IT specialists, doctors, and senior management personnel who had an average period of work experience at the hospital of three years.

We adopted a practical method to understand the textual data that was suggested by

Klein and Myers (1999). It involved personal visits, emails, and telephone calls to foster good relationships among the researchers and the key informants. It enabled us to understand the rich descriptions (Yin 1994), thus better revealing the meaning of the participants' responses. This enabled us to identify the core case information necessary to comprehend, analyze, and evaluate the data (Klein et al. 1999). To ensure sound quality of the data collection, we triangulated the data with other resources, including empirical observations and follow-up email clarifications, along with about 230Mb of softcopy documentation and archival records.

3.2. Data Analysis

Data analysis was performed in tandem with data collection; Eisenhardt (1989) suggests that this is appropriate for case research. Based on the agility, IT innovation, and innovative-capability literature, we identified an initial set of themes including sensible agility, compliant agility, interactive agility, transformative innovation, interactive innovation, and adaptive innovation. These are seen as pertinent to agile, innovative-capability development. Based on these themes, we further refined the theoretical lens guiding the analysis, and we searched for other themes emerging from the data (Eisenhardt 1989). The data were organized and coded according to these themes, and each new finding was verified from at least two other sources of data (Klein et al. 1999). With the emerging themes, preconceptions were addressed through logical reasoning and via the data-triangulation method (Klein et al. 1999) in order to challenge the existing schema (Walsham 2006). Data were then verified by comparison between the empirical data, the theoretical lens, and the findings drawn from relevant literature. This enabled us to sharpen the construct definitions and to enhance the theoretical explication of model development (Eisenhardt 1989).

4. THE CASE STUDY

4.1. Case Background

Health IT has become a priority for the Singapore government since the 1999 restructure of its healthcare providers into two vertically integrated clusters: the National Healthcare Group (NHG), and Singapore Health Services (SingHealth). The call to “exploit IT maximally” by the Minister of Health of Singapore further encouraged hospitals to improve their healthcare systems. Alexandra Hospital, one of the poorer performing hospitals with high levels of patient complaints, was challenged by Health Minister Khaw in February 2004 to set up a new healthcare standard and to deliver a “hassle-free”, patient-centric hospital by 2009 (Alexandra Hospital 2009).

The challenge from Minister Khaw triggered AH management to be strategic in focusing specifically on IT-based improvements. For years, AH employed traditional management methods resulting in poor patient service and long waiting times. In the emergency ward, at times patients would have to queue for three or four hours before being seen by a doctor (not accounting for the time taken to be warded properly). The situation worsened, as shortages of medical staff increased. The morale among medical staff was low owing to overwork and insufficient IT support. A senior patient-service associate (PSA) highlighted the situation:

We managed the beds by using mini cards; when a bed is ready for cleaning, a card will be placed at the counter; however, because there are always so many papers on the desk it is hard for us to trace them; we just have to manage in chaos.

AH lost its competitiveness to its rivals through losing patients and incurring time losses. In response to the crisis, AH management became more strategic and they benchmarked the hospital to become the “Mayo Clinic of Asia”, complementing this objective with the aim of providing patients with a “hassle-free”

experience. To counter the crisis of poor performance, many strategic initiatives were adopted at AH. This resulted in it eventually being voted the best-performing hospital in the Ministry of Health Patient Satisfaction Surveys for six consecutive years—2004 to 2009 (Alexandra Hospital 2009). The Director of Operation reflected on AH's survival:

Because we are small, we do not have the critical mass, therefore we are slack to do things. However, because we are small, it allows us to move faster ... 'Abundance makes you poor' ... Out of necessity, we become creative; we invent new ways of doing things to compensate for our lack of resources. To us, it's survival.

4.2. Phase 1: Grounding for a Strategic, Innovative Breakthrough

AH objectives centered on searching for the fastest and most efficient ways to provide a unique patient-centric service. The hospital adopted a unique combined approach. First, they drew on the best practice of US and Japanese medical and (or) healthcare systems (the Mayo Clinic in the United States, and the Kameda Medical Centre hospitals in Japan) on four key aspects. These were to match and exceed the best performers; achieve the lowest infection rate; have the shortest length of patients' stay;

and to achieve the lowest average bill. They integrated these with the Japanese "kaizen" (continuous improvement) philosophy, taken from the Toyota (manufacturing) plant in Japan and its workshop in Singapore. The AH management team exploited their strong collaborative culture between managerial and clinical staff to win over the relevant stakeholders, before commencing a series of kaizen activities (see the references to sensible agility capability in Table 2).

It was not easy to reach consensus in simplifying clinical and managerial processes among the group of seven operational team members who were supported by clinical staff. Thus, the hospital always reverted to its "patient-first concept" to resolve conflicts. The Director of Projects and Operations noted:

It isn't [that] the doctor who shouted loudest would get his idea adopted, but it is through a common understanding, whereby one's idea could provide the best solution for patients.

In addition, the strategy of building on the collaboration between managerial and clinical staff improved the innovative capability of different stakeholders working together to scrutinize traditional work processes, identifying bottlenecks, reviewing the existing patient-care processes, and in re-examining patient services

Table 1. Number of interviews conducted with different hospital stakeholders

Informants	Number of Interviews	Number of Follow-ups	Total
Director of Projects and Operations	1	3	4
Assistance Director of Projects and Operations	1	3	4
IT consultant	1	2	3
Project Specialist Operations	1	6	7
Senior Doctor	1	1	2
Head of nursing	1		1
Senior nursing officer	1		1
Senior Patient Service Associate	1	1	2
Total			24

Table 2. Capability development in achieving an agile-innovation strategy in Phase 1

Sensible Agility Capability	
Sensible towards compulsion for improvement	"You have to be sensible to our surroundings; you cannot hang on to the old horse, the old technology. You must leapfrog, move onto a new platform by learning from others. That's why we were sent to the US and Japan." Director of Projects and Operations
Sensibly respond to match customer needs and organizational resources	"We try our best to map the patients' 'wish-list' within our aim, capacity and resources." Assistant Director, Operations "With this information we have to reasonably design a flexible system where it allows us to have a good match of supply and demand, the waiting time simply because there isn't a good matching of supply (doctors) and demand (patients) situation." Assistant Director of Projects and Operations
Transformative Innovative Capability	
Innovative capability to transform and improvise concepts and knowledge	"It worked for Toyota cars, so why not for patients at hospitals?" A management team member "You cannot wait for people to tell you but you can learn from them and improvise. In our case, we have to be the first mover and tell people how we want the system to perform, so there's a lot of visualization needed." Director of Projects and Operations
Innovative capability to collaborate for transformation	"The contribution is from everyone because the process flow and the initiated changes have to happen on the ground then we (the management) help to 'systematize' the components or the requirements." Director of Projects and Operations

via the use of a kaizen flow. For example the traditional nurse-call system was less flexible, so that nurses may have needed to make a few trips to obtain the amenities they required. According to the senior nursing officer:

Now our patients can speak to us through the intercom. That allows us to pick up amenities as required. We have cut down unnecessary trips. This new process is improved by the use of technology.

Adopting an approach based on the idea of kaizen, management wanted to ensure that staff at AH could acquire the ability to collaborate, and the ability to provide speedy feedback in identifying various problems at the point of kaizen (between departments), and process kaizen (within department). In this regard, the Director of Projects and Operations noted:

We are all part-timers; working above all this we have regular jobs ... so to be able to identify problems within two months, I would say we are 'LEAN'.

Basing the strategy on a patient-centric mindset, weekly system-review sessions were organized to gather suggestions and feedback from all stakeholders, including patients. Within a three-month review period, stakeholders and management identified certain critical areas that needed to be changed, including the Department of Emergency Medicine (DEM), the Bed Management Unit (BMU), and the nurse-call system.

In supporting the transformation through consultations with volunteers, the operations team discovered that overlapping activities existed in their operating procedures. To overcome this, the clinical staff gathered to test the system flow (the system kaizen concept). They were encouraged to suggest changes which would be channeled to the monthly feedback session chaired by the CEO, Chief Operations Officer, or the Chief Medical Officer. According to the Assistant Director of Operations:

They will remind us not to tell them about good news, but to tell them the bad things or bad news, tell them anything that we don't like.

A second strategy was initiated to upgrade system responsiveness for patients, with a target of a “zero” waiting time for patient admission via DEM, and a lead time of 10 minutes for all patients from the DEM call to the arrival at the designated bed. Then, AH management further benchmarked its patient service with other service-industry standards, such as the hotel and airline industries. To increase creativity among the staff, monetary incentives were offered to stimulate more efficient and effective innovative ideas. A test-run method was introduced to examine the proposed ideas for reflection and improvement before conducting the system walk-through to eliminate unnecessary processes to meet the DEM and BMU users’ requirements.

The operations team encouraged IT vendors to explore the potential value of their service-oriented architecture to integrate the bed-management system and the nurse-call system in a common platform. According to the Director of Projects and Operations:

Our boss is trying to champion the idea, if not now, for the next three to five years; we still need to change. So why not take the lead if necessary? If not we are just going to follow. Whatever it is, sooner or later you still need to revamp and change again. Why not look at it seriously, have an eye to the future and the future good?

With this futuristic leadership, AH encouraged the clinical and managerial staff to transform and automate their work practices innovatively. Table 1 contains references to such activities.

4.3. Phase 2: Retrofitting for Strategic Innovative-idea Integration

Once the initial strategic changes were in place at AH, and the combination of processes and technologies was selected, the operations team proposed the establishment of a common operational platform with integrated healthcare

information systems. Following numerous cost-benefit analyses, management decided to invite external collaborators to form strategic partnerships to improve the physical layout, business workflow, and the integrated healthcare information systems of DEM. The Director of Projects and Operations explained:

We are a small hospital. So [for] some of the things we implemented, cost is important, as we don’t have the financial arm to bring in the big software players.

AH sent the proposed DEM workflow requirements to a Singapore polytechnic college—as an assignment topic—in an attempt to obtain a free HIS prototype. The Assistant Director of Projects and Operations explained:

To be able to solve an actual business problem is a precious opportunity for those polytechnic students.

Within the 90-day commissioning period, AH received the IS prototype for DEM with a newly designed system. Taking this prototype from the polytechnic students allowed AH to advertise the students’ achievement as bait for IT solution-providers to come forward with a better solution. Eventually, Fujitsu and Cisco joined forces to improve the system further. After several test runs based on the prototype, a problem was identified. For the hospital to provide a “hassle-free” DEM visit, patients’ waiting times had to be reduced still further. Building on the operations team’s ability to identify issues through a series of kaizen activities (see Table 2 for references to the vigorous and vigilant AH attitude to solving issues), a pull system was suggested as the solution. According to the Project Specialist, Operations:

When you introduce a pull system for beds, it means that you are providing information to the DEM staff on which beds are available for disposal, rather than having the necessity for nurses calling to look for any available bed.

Table 3. Capability development in achieving the agile innovation strategy in Phase 2

Harmonize Agility Capability	
Vigorous in confronting issues	“This is the culture I would say more or less encouraged and influenced by the CEO and COO. We look at things, if it doesn’t make sense, or is getting difficult, then we look for ways to improve it.” Project Specialist, Operations
Vigilant in solving problems	“When someone pointed out an error, we review it quickly but carefully. I always tell them I only have five minutes to tackle some of the issues immediately, so it became immediate. Don’t slip away, it just not solving the problem, we want to be quite prompt.” Director of Projects and Operations
Flexible in accommodating changes	“We designed the architecture in such a way that we decouple the data, decouple the processes, yet seamlessly try to link them together, so if we want to change processes, just change. The model will not break apart, so this finding of our definitive moment for all of us to change healthcare.” Director of Projects and Operations
Integrative Innovation Capability	
Integrate and transform the new ideas into actionable plans	“Out of necessity, we become creative; we invent new ways of doing things to compensate for our lack of resources. It immediately drives our necessity. To us, it is survival.” Director of Projects and Operations “Our learning is that, by introducing technology too early, you might end up with very interesting phenomena. For example, we recently launched the wireless ward system at the Bed Management Unit, where we focused on the turnaround of the beds. So we work backwards with our partners to focus on the process, like how technology can improve the process.” Director of Projects and Operations
Retrofit the systems tactically with improvised ideas	“We are talking about retrofit, now that we have the infrastructure in place, we are thinking how to innovatively use the Nurse Call System. By pressing the button, patient can talk to the nurses, so that on the way they can pick up the amenities as required.” Director of Projects and Operations “We use the Cisco system as a whole but in addition, and we build BMU awareness, therefore it then becomes the bed awareness system. So it is now different from the original purposes of Cisco’s system.” Project Specialist, Operations

To meet the need for a pull system, AH extended its partnership with Cisco to bring in GlobeStar and Fujitsu—who provided the hardware support for the systems. The hospital integrated the functions of the DEM and BMU systems according to AH user requirements (see Table 3 for references to how AH retrofitted these systems). In working with the external parties, the operations team provided ideas gathered from users to assist the pilot test for the DEM, and organized the integration of a wireless device which updates a database that keeps track of patient admissions and discharges (see Table 3 for references to the hospital’s flexibility in accommodating changes). This technology was tested and rolled out to support

the operational flow. When a patient was ready for discharge, the ward PSA updated the database system to the “planned/intended discharge” function, so that a task-request message was sent to housekeepers; beds would be ready within three hours. Simultaneously, another message was sent to the DEM to prepare the patient for immediate discharge. This synchronization and the integration of activities, workflow, and processes between departments were tested in Ward 13 (see Table 4 for references to how AH integrated and transformed new ideas into practice). Significant improvements were achieved with the integrated systems and with the wireless devices. This ward became the prototype for the rest of the hospital.

Table 4. Capability development in achieving an agile innovation strategy in Phase 3

Interactive Agility Capability	
Emphasis on providing interactive support to clinical staff and patients	"After the test, the scalability and the follow-up become crucial. Meaning, once it is successful, how to quickly scale to the rest [of the wards], so that the diffusion would be spread throughout the organization." Director of Projects and Operations
The interactive and flexible function provided by the system	"I enjoyed the stateless system support, where it gives me the flexibility to change the status of the beds, unlike other tested system which forced us to repeat the entire workflow if we were to amend the bed status." Project Specialist, Operations
Adaptive Innovative Capability	
Ability to learn and adapt for improvement	"Just within a few weeks, we hardly hear from Ward 13 on any problems, complaints or things that cannot be done in time; in fact, everyone started sending patients to Ward 13. In fact, everyone is highly commended about Ward13." Project Specialist, Operations
Foster innovative thinking and practices	"These days, the switch to the computerization has quickened the process of staff adaptation. When it comes to systems and computers, I think Singaporeans are fast. The introduction of new technologies are seen as 'new toys' for the boys and we enjoy it and we want more!" Senior Doctor from DEM

4.4. Phase 3: The Founding of the Strategic Innovative Healthcare Information Systems

After the successful piloting of the new strategies within the DEM and BMU, the systems were fully installed across the hospital. To improve the interactivity and agility of the systems further, the emergency ward was renovated to ensure that point-to-point stations were suitably located to facilitate effective and efficient services. After a careful review, AH staff identified another bottleneck in DEM (see Table 4 for references to AH interactive agility capability). Thinking laterally and giving priority to serving the patients better, the head of DEM revamped the traditional triage system by assigning an experienced senior doctor instead of a junior doctor (trainee medical officer) to act as the first point of contact for patients at their initial consultation. This move at first raised doubts about the wisdom of the decision; it later proved to be a successful strategy because—combined with the new IT-driven processes—workflows and patient movement improved (see Table 4 for references to the hospital's adaptive innovative capability). This change enabled AH to explore a new paradigm for service delivery.

As part of sustaining the success of these changes, briefings were routinely scheduled to update the clinical staff and to provide consistent support in implementing the new arrangements. According to the senior nurse manager:

Kaizen is a journey of improvements. We are still striving for perfection through continuous improvement and maintaining consistency.

The operations team designed a user-friendly system. Less than a week after the roll-out, the PSAs could use the system without any problems.

Taking a bold initiative to improve the workflows and innovatively developing a better system from scratch is a real achievement and was considered a significant strategic outcome for AH. The entire innovation service-transformation was successful: AH improved patient satisfaction, and successfully expedited patient flow, on average, from 22 to 70 patients attended to per hour; AH reduced patient waiting-times by 50 per cent; and the newly designed system was considered to have contributed genuinely to the recording and retention of patients' medical histories via a new paperless form of administration. As a result of

Table 5. *Capability development in achieving an agile innovation strategy for BMU*

A Process Model of Strategic Agile Innovation Development for BMU		
Phase 1: Grounding for a strategic innovative breakthrough	Phase 2: Retrofitting for strategic innovative integration	Phase 3: Founding of the strategic innovative Healthcare Information System
The Agile Capability Process		
Sensible Agility	Harmonizing Agility	Interactive Agility
“For many hospitals, the patient would have to wait for a bed for a few hours. To be quick, we sensibly reference to the previous success executed by Toyota Production concept- just in time...borrowing from that concept, we explore the ‘pull system’.” Commented the Project Specialist of Operations.	“Good working relationship is important, that’s why we always trying to resolve and improve the communication between the wards and the various departments, and various teams of nurses so that they can work closely with each other when the patient is going through the discharge process” Commented the Project Specialist of Operations.	“Cisco provided us a communication system to test our concept. With this if the doctor called for planned discharge, the system will broadcast the message so the nurses can prepare for Medical certificate, discharge notes, financial building, notify patients family to pick-up the patient as well as getting the medication ready for them to bring home...this has improve efficiency and enable respective nurses to response almost immediately.” Assistant Project Specialist of Operations
The Innovative Capability Process		
Transformative Innovation Capability	Integrative Innovation Capability	Adaptive Innovation Capability
“We like to use the term kaizen in Toyota production. It improves on the work procedures. Sometimes, some kaizen for procedures save about 1 or 2 minutes. Over the long run, you save quite a lot of time. That is why I always ask we nurses to share their ideas to keep their processes lean...” Head of nursing	“We chose the ward that statistically shows the faster turn rate, because that implies that the nurses in that ward work harder and have good cooperation with other medical departments and wards... thus would be more ideal for us to test the new system.” Director of Projects and Operations	“For the BMU, we made the system simple so that it is not so cumbersome to update.. It is pretty simple and they are quite techno savvy. So, what we do is show them a few times then let them teach themselves. For this BMU, nurses just need to recognize the status and change it accordingly. Pretty simple...” Head of Nursing

the key strategies implemented, AH boosted its brand name in the hospital industry.

Table 5 shows some snapshot issues relating to agile and innovative capabilities that smoothed the progress of the BMU development.

5. DISCUSSION

Based on this AH case study, we developed a process model (represented in Figure 1) to explain how agility and innovation capabilities were strategically nurtured, developed, and managed by a private hospital in setting up a

new hospital-industry standard for Singapore. Our process model suggests that an agile innovation strategy entails a complex helix process of a firm’s sensitivity and responsiveness to integrating and reconfiguring resources to cope with innovative changes, both within (Eisenhardt et al. 2000) and outside the hospital (Wang et al. 2007). Given that the model is inductively derived from the AH case study, our discussion provides an explanation of how the existing literature confirms the model and how the model enriches existing perspectives on HIS agile innovation.

Figure 1. A process model of strategic agile innovation development

A process model of strategic agile innovation development		
Phase 1: Grounding for a strategic innovative breakthrough	Phase 2: Retrofitting for strategic innovative integration	Phase 3: Founding of the strategic innovative Healthcare Information System
The agile capability process		
Sensible agility	Harmonizing agility	Interactive agility
To establish a sensible position from which to fulfill external (c.g. government) and internal (c.g. patient) expectations through the exploration of internal operations in order to improve innovation and competitive action - Sensible approach towards compulsion for improvement - Sensibly responds to match the customer needs and organizational resources	To fit and match external and internal requirements, practices and expectations exclusively in enriching the patient experience - Vigorous in confronting issues - Vigilant in solving problems - Flexible in accommodating changes	To provide interactive support and collect inventive ideas for the continual improvement in designing flexible systems with high interactivity functions. - Emphasize on providing interactive support to clinical staff and patient - Continue improving the interactive and flexible system's functions
The innovative capability process		
Transformative innovation capability	Integrative innovation capability	Adaptive innovation capability
Enhanced internal and external resources in conjunction with collaboration between managerial and clinical staff to innovatively transform the HIS clinical processes - Innovative capability to transform and improvise concepts and knowledge - Innovative capability to collaborate to implement transformation	Articulate to integrate technologies, managerial, clinical processes, and resources from partnerships to develop an innovative patient-centric healthcare systems - Retrofit the systems tactically with improvised ideas - Integrate and transform new ideas into actionable plans	Facilitate users' system adaptability and welcome innovative suggestions to continue improving systems function. - Ability to learn and adapt for improvement - Foster innovative thinking and practices

5.1. Phase 1: Grounding for a Strategic Innovative Breakthrough

With the goal of becoming a leader in the healthcare industry, AH was initially driven to improve its IS systems by implementing significant changes, overcoming thorny challenges and sourcing problems, as well as satisfying external (government) and internal (patient) expectations before establishing an innovative strategic breakthrough. The Director of Operations commented that:

In a very dynamic and chaotic environment, strategy-making to us means strategic road-mapping. It is very hard for people to manage it well. All the more, it's an abstract-looking innovative new way of operation, making it simple for the ground staff to operate. To get

all these concepts and processes to work out well, we have to idiot-proof them ... and I'm not saying that our doctors like that.

A two-sided approach—based on an analysis conducted by senior management through sensible agility capability—was adopted by AH to implement their new strategy. That is, the capability to demand improvement, and to ensure that customer needs and organizational resources were matched. Sensible agility capability meant AH responding to the internal and external changes by mapping their strengths and weaknesses with its “patient-first concept”. For example AH was sensible enough to respond by launching its strategic innovation plan to match patient needs with organizational resources and its new strategy—the “patient-first concept”. They carefully examined the fit between the

organizational system—including a management system, an information system, and a healthcare system—and its environment (Heeks 2006). This managerial effort is worth mentioning, as it played a critical role in the success of the HIS designs (Kaplan et al. 2009). With such initiatives, AH took the stance of a first mover with a manageable IT investment risk.

Equipped with sensible agility capability, the hospital further transformed itself based on the empirical evidence uncovered at AH with two crucial steps introduced to foster its innovative IT capability. These were their capabilities in transforming and improving concepts and knowledge, and in collaborating on driving the transformation. As evident from this case, AH demonstrated an ability to articulate knowledge by transforming concepts, and extending knowledge gained from its overseas counterparts to revolutionize its healthcare workflow processes. Second, with the introduction of the kaizen concept, the hospital further enhanced collaboration among hospital stakeholders by mapping the patient-centric strategy with service-oriented architecture to integrate the bed-management system and the nurse-call system. They were eventually able to mitigate operational-process bottlenecks and improve the glitches in the workflow which contributed to high patient waiting-times and poor patient services. Inevitably, these capabilities transformed IT innovation, reorganized healthcare processes, and facilitated innovative breakthroughs that resulted in an effective, measured improvement in the management of patients.

5.2. Phase 2: Retrofitting for Strategic Innovative Integration

In the second phase of HIS design, AH strategically had students as the “fresh pair of eyes” in designing their HIS prototype. To enhance the design further, AH faced the challenge of needing to integrate and articulate the ideas offered by external (partners) and internal (managerial and clinical staff) stakeholders. AH did face some of the serious “design-reality gaps” (Heeks (2006) between the new system’s design requirements

and the realities of the resources available within the hospital. This applied too between the new processes conceived within the system design and their existing processes. To harmonize the situation in an agile manner, AH focused on retrofitting for strategic innovative integration by introducing three capabilities. These were vigor in confronting issues; flexibility in accommodating changes; and vigilance in solving problems. These capabilities enabled AH to lessen the tension between internal and external stakeholders through vigorously confronting issues before accommodating changes. For example The Project Specialist of Operations shared her challenge:

Sometimes it is a little bit erratic; the external consultant will ask me: ‘Where are all the doctors?’ ... The external consultants would expect the doctors to attend all the meetings. However, most of the time, doctors were held up ... to tell you the truth, we cannot make doctors attend every meeting. We have patients who need emergency treatments ... this is the tension that I have to deal with between the different stakeholders.

Confronting issues involved leveraging and matching resources—such as skills and knowledge—to respond with vigilance to expected and unexpected events effectively, by exploiting any opportunities that arose (Mathiassen et al. 2006; Sambamurthy et al. 2003). An additional example that illustrates flexibility in accommodating changes was shared by the IT consultant.

It is a matter of getting used to the system, because we keep telling them, don’t worry, this is just the state of the bed. If it requires changes, we will program it for you ... this is part of the Cisco program, the behavior of the system is stateless. Thus, it provides lots of flexibility for us to adjust and fine tune.

However, flexibility comes with the need for watchfulness. To resolve problems and to

ensure patient safety, AH retrofitted innovation with strategy. The Project Specialist of Operations commented.

To crystallize all the issues that we collected, first the operations team will meet anytime we are free to filter all the issues before involving other members for brainstorming ... for technical issues, we meet the technical group every Monday night. Every fortnight, my team members and other members join this WOW meeting, sharing and listening to what our patients need, so that we can incorporate their needs to build a hassle-free system.

As shown in this case study, AH successfully created complex and new-to-market innovations through its agile harmonization capability in responding to conflicts, and by exploiting partnership opportunities (Koch et al. 2008). According to the Project Specialist of Operations:

We have no problems with people who resist this implementation. When people resist, it means that there must be some problems with the system. So we have to stop for a short while and go manual ... Of course I cannot say it's plain sailing ... We have some problems along the way. So there are times when we need to stop and fix all the problems before we can proceed ...

In particular, harmonized agility nurtured hospital's innovative capability to integrate technologies, managerial and clinical processes, and resources provided by partners. This integrative innovation capability was made possible by two means: retrofitting the systems tactically with improvised ideas; and integrating and transforming new ideas into actionable plans.

First, with strong senior-management support, AH retrofitted the systems tactically by investigating new avenues to modify and improve current processes. Such capabilities enable an organization to access, process, and interpret the information needed to resolve

complicated problems and (or) issues in a purposeful manner (Glynn 1996; Lawson et al. 2001). In this case, the hospital could increase its potential to adapt and operate strategically. Second, the hospital's ability to integrate and transform new ideas was shared among different stakeholders, who molded them into actionable plans that they developed by learning new mechanisms—including continual experimentation and the analysis of mistakes—described as “organizational intelligence” (Eisenhardt et al. 2000). These abilities are critical to the success of innovative systems and process transformation, as is evident in the case of AH (Andersson et al. 2005; Cohen et al. 1990).

5.3. Phase 3: The Founding of the Strategic Innovative Healthcare Information System

The founding of the strategic innovative HIS was achieved in the last phase with the application of two new systems at AH over a period of 18 months. During this phase, more emphasis was placed on agile interactive support to users, especially in responding to their requests in relation to fine tuning the systems. In this process, AH applied “outside the box” thinking through two agile interactive capabilities: providing interactive support to clinical staff and patient feedback (soft system); and continuing to improve the designs of interactive and flexible system functions (hard system). It was evident in this case that the interactive support provided by the AH operations team encouraged users to utilize the new systems, but it also nurtured the team's adaptability and flexibility (Sherehiy et al. 2007) in predicting changes and responding to unanticipated requests quickly and efficiently (van Oosterhout et al. 2006). As a result, this capability led to better synergy in the founding of the interactive HIS. After implementation, this innovatively designed system offered medical staff interactive and flexible functions in accessing accurate real-time information to deliver a better quality of patient care (Sallas et al. 2007).

Grounded in the empirical data, our model suggests that the processes of innovation and continual improvement benefited system users, and also the operations team at AH. To “explore a new paradigm for service delivery” in this phase, two adaptive capabilities were developed to close the loop in the agile innovative process. These are the ability to learn and adapt for improvement; and to foster innovative thinking and practices. As was revealed in this case study, the ability of the operations team to learn and adapt to user requests created knowledge (Teoh et al. 2009) and organizational intelligence, which in turn fostered innovative thinking and practices within the hospital. These capabilities empowered AH to continue improving its HIS and to rank consistently as the best-performing hospital in Singapore in providing hassle-free, patient-centric services (Alexandra Hospital 2009).

6. CONCLUSION AND IMPLICATIONS

According to the World Health Statistics 2014, the total expenditure on health as percentage of gross domestic product grew from 8.2 to 9.1 from 2000 to 2011, and the per capita total expenditure on health (at average exchange rate, US\$) grew from 485 in 2000 to 1007 in 2011. For Singapore, the per capita total expenditure on health (at average exchange rate, US\$) was 662 in 2000 and 2144 in 2011. While Singapore enjoys relatively high ranking of the world's health system (e.g. Life expectancy ranks among top 5) (World Health Organization 2014), along with its potential to transform societal health and wellness levels, Singapore's Ministry of Health has devoted to develop the world's most cost-effective healthcare system. It is recognized that health care organizations must continue to invest significant capital in information technology. These investments in health information technology are essential to comply with government mandates. The investment in IT has great potentials to improve healthcare services in many ways, our study

suggests that investing in ICT-enabled agile innovative capabilities might be an effective way to improve healthcare services.

The purpose of this research paper is to enhance current understanding of strategic information systems by conceptualizing the generation of agile innovative capabilities in the healthcare sector. The process model proposes strategic descriptions that analyze how agility and innovation capabilities were nurtured and managed during the implementation of healthcare IT in a hospital setting, with the aim of establishing a milestone for the healthcare industry in Singapore. As is evident in this case study, our proposed model depicts the strategic execution of agility and innovation capabilities and we examined this in three phases: grounding for a strategic innovative breakthrough; retrofitting for strategic innovative integration; and founding a strategic innovative HIS. In addition, the findings substantiate the notion that knowledge of how agility and innovation capability can improve the quality of healthcare services remains weak. This needs to be addressed, especially since attaining agile innovation in the healthcare industry remains a particular challenge (Harrington et al. 2010).

Although Healthcare IT is regarded as a necessity by prior research and reports, the strategies to achieve implementation and adoption of healthcare IT for better service quality and performance might vary in different countries, given the differences in financial, technological, and human resources (Bagayoko et al. 2010; Heeks 2006). Based on an ongoing, large-scale action research project within the health care sector (called HISP) in a number of developing countries, Braa et al. (2004) identified two broad themes that underlying unsuccessful attempts with HISs in developing countries, including challenges to make an information system work over time and the problem of how to make one, working solution spread to other sites, and be successfully adapted there.

Taking all these possible challenges on board, our research focuses on a healthcare system implementation case of a small hospital in a developed country, but with limited

resources to bring in the big software players so they chose to start from a prototype from polytechnic students. The three-phase implementation process started with relative small financial investment but achieve good outcome at the end with well-controlled budget. This is a successful case study as it is rigorous and consistent organization of the actors and processes of care and a clear choice for the establishment of infrastructure (hardware and software), while the financial burden was also carefully managed (Bagayoko et al., (2010).

“Clever technology can help solve two big problems in health care: overspending in the rich world and under-provisioning in the poor world.” (The Economist, April 18, 2009). Our findings suggest that, the strategic execution of agility and innovation capabilities could be achieved through the proposed three-phase process model, with reasonable cost. Just as important, our research findings could have potential implications for healthcare industry in not only developed countries but also developing countries.

For a future study, it would be interesting to apply this model to explore the “design-reality gaps” more deeply from the agility and innovation capabilities perspectives in healthcare settings; we found that this was a constant challenge (Harrington et al. 2010). It would also be interesting to substantiate findings further through a quantitative study of the notion of how agility and innovation capability can improve the quality of healthcare services. This might include extending this study beyond the dynamic capability theory’s perspective.

6.1. Theoretical Contributions

This case study highlights implications for researchers and professionals. Theoretically, the model proposes and explains a series of structured activities and systematic processes, learned from the case study, that aim to develop strategic, agile, innovation capabilities. The model contributes by providing some answers to fill a gap left by the lack of innovation studies in the context of a healthcare sector that

involves certain types of service firms (Aranda et al. 2002).

While a large part of prior research on agility, innovation, and dynamic capabilities focuses on manufacturing firms and supply-chain management in large organizations (Dobrzykowski et al. 2014; Yusuf et al. 2004; Yusuf et al. 2014), our research complements the current literature on strategic, agile, innovation capabilities in a safety-critical industry—healthcare. The healthcare industry represents an important and growing research context that can be characterized by an intense focus on providing individualized offerings and a criticality of quality and safety in delivering patient care (Dobrzykowski et al. 2014; Harrington et al. 2010). Thus, it calls upon critical, emergent needs to develop agile innovation capabilities in the healthcare industry. In this research, a patient-first concept, a service-oriented architecture, and a service-delivery paradigm are the three additional variables that are empirically derived from our data and which are specific to the healthcare service industry. These three variables are different from the original theory and may not be applicable in, say, manufacturing industry.

This study identified two key variables to postulate radical innovative capabilities study by Subramaniam and Youndt (2005), they are (1) “fresh pair of eyes” and (2) “outside the box thinking” empirically derived from the case study. These two key innovative variables clearly contributed to revolutionize Singaporean healthcare delivery. This can be seen in the recognition given by the Singapore Health Minister that AH is the key prototyping center which leads and sets future standards for the Singapore healthcare industry (MOH, 2010).

Normally in conducting a single case study, the common criticism is whether the research can be generalized (Walsham 2006). We assert that this case study is valid and can be generalized within its context; that is, in and across developed nations’ hospitals, as it invokes the principles of “analytic generalization” (Yin 2003, p.32). That is known as “generalizing from description to theory” (Lee and Baskerville,

2003, p.235). This derived model is grounded in the empirical reality of our case study, and it is supported by existing literature. Thus, constructs identified in the proposed process model (Dyer and Wilkins 1991) could be extended for statistical validation, so that the boundary conditions of our study can be refined and made applicable across developed countries for use as a guide for emerging or developing nations.

6.2. Practitioners' Contributions

For practitioners, this study contributes three key insights in relation to strategic planning, cultivating, practicing, and managing an agile, innovation capability in the healthcare industry. Managers need to be sensible about their goals and plans, and persuade clinical and managerial staff about proposals before setting up internal and external collaborations. Managers must be flexible in their provision of support to clinical and managerial staff, especially in soliciting and matching skills with the available technologies. This is because aligning clinical and managerial staff requirements remains a critical challenge for hospitals. Finally, managers must be responsive in providing interactive support to users, and in encouraging suggestions for the continued improvement of the system's functions to provide a higher quality of care and service to customers.

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ENDNOTES

- ¹ See the home page of the website of the Ministry of Health, Singapore (www.moh.gov.sg) [accessed September 25, 2010].